

Data Structures and Algorithms - Assignment 4

1. Write a linear search algorithm to return index of last occurrence of key.
2. Implement linear search algorithm to find the nth occurrence of the given element. If nth occurrence is not found, return -1.
3. Implement binary search algorithm if array is sorted in descending order

Optional

4. find the first non-repeating element: Input: { 1, 2, 3, -1, 2, 1, 0, 4, -1, 7, 8 } Output: 3
5. to find rank of an element in a stream of integers. rank: rank of a given integer "x", in stream is "total no. of ele's less than or equal to x (including x)".
 - Input: { 10, 20, 15, 3, 4, 4, 1 }
 - Output: Rank of 4 is: 4
6. <https://leetcode.com/problems/remove-all-occurrences-of-a-substring>
7. <https://leetcode.com/problems/missing-number>
8. <https://leetcode.com/problems/remove-element>